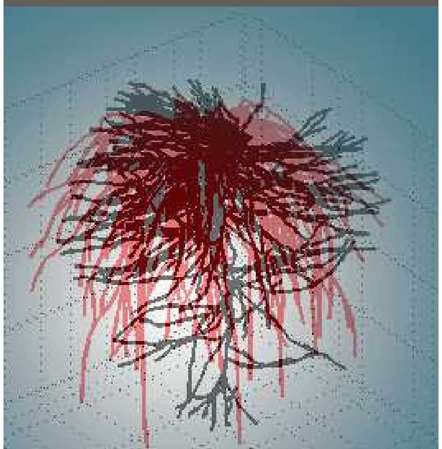
Published November 14, 2013

Special Section: VZJ Anniversary Issue

Mathieu Javaux*
Valentin Couvreur
Jan Vanderborght
Harry Vereecken



A critical review of implicit assumptions behind typical one-dimensional root water uptake and stress models is performed. Plant-scale mathematical expressions for stress and uptake functions obeying Darcy flow equations in three-dimensional systems are proposed.

M. Javaux, J. Vanderborght, and H. Vereecken, Institut für Bio- und Geowissenschaften, Agrosphäre (IBG-3), Forschungszentrum Juelich GmbH, Juelich, Germany; M. Javaux and V. Couvreur, Earth and Life Institute, Université catholique de Louvain, Louvain-la-Neuve, Belgium. *Corresponding author (mathieu.javaux@uclouvain.be).

Vadose Zone J.
doi:10.2136/vzj2013.02.0042
Received 10 Feb. 2013.

Root Water Uptake: From Three-Dimensional Biophysical Processes to Macroscopic Modeling Approaches

The process description of plant transpiration and soil water uptake in macroscopic root water uptake models is often based on simplifying assumptions that no longer reflect, or even contradict, the current status of knowledge in plant biology. The sink term in the Richards equation for root water uptake generally comprises four terms: (i) a root resistance function, (ii) a soil resistance function, (iii) a stress function, and (iv) a compensation function. Here we propose to use a detailed three-dimensional model, which integrates current knowledge of soil and root water flow equations, to deduct a one-dimensional effective behavior at the plant scale and to propose improvements for the four functions used in the macroscopic sink term. We show that (i) root hydraulic resistance may be well defined by the root length density but only for homogeneous lateral conductances and no limiting xylem conductance—in other cases a new function depending on the root hydraulic architecture should be used; (ii) soil resistance cannot be neglected, in particular in the rhizosphere where specific processes may occur that alter the soil hydraulic properties and therefore affect uptake; (iii) stress and compensation are two different processes, which should not be linked explicitly; (iv) there is a need for a clear definition of compensatory root water uptake independent of water stress; (v) stress functions should be defined as a maximal actual transpiration in function of an integrated root–soil interface water head rather than in terms of local bulk water heads; and (vi) nonlinearity in the stress function is expected to arise if root hydraulic resistances depend on soil matric head or when it is defined as a function of the bulk soil water head.

Abbreviations: RLD, root length density; R-SWMS, Root–Soil Water Movement and Solute transport platform model; RWU, root water uptake.

Evapotranspiration returns as much as 60% of the whole land precipitation back to the atmosphere (Oki and Kanae, 2006). Through transpiration, plants act as a natural regulator of water fluxes between the soil and the atmosphere. The transpiration fluxes depend on environmental conditions such as rainfall and atmospheric controls but also land use practices and agricultural management. These environmental conditions are strongly impacted by the ongoing climate change leading to increasing temperature, changing rainfall patterns and intensities, and an increase in the frequency of floods and droughts. Besides being impacted by weather and climate, there is also an important feedback of plant transpiration on weather, climate, and its changes. Plants, with their capacity to take water from the soil, play a key role in land surface–atmosphere interactions, which are important to predict extreme events such as droughts and heat waves (Seneviratne et al., 2006). Teuling et al. (2006) demonstrated that uncertainty about the modeling of root water uptake in land surface schemes may lead to an important uncertainty about the evolution of plant transpiration during dry spells, which has of course important consequences for the feedback with climate predictions.

With the current global challenges for food and feed production but also for maintaining ecosystem services, optimizing green water use and securing and increasing plant productivity have become central in the current social concerns (Vereecken et al., 2009). Modeling how plants react to drought in the short and long term is crucial to be able to mitigate future threats to plant production for food, fuel, and fiber (Betts, 2005). Understanding how plants regulate water flow under changing environmental conditions is therefore an important scientific question (Jury et al., 2011).

Today there is a general consensus on the tension-cohesion theory to describe the ascent of water in plants. This theory states that that the water is passively extracted from the soil and flows to the atmosphere through the plant. The catenary hypothesis (Van den Honert, 1948) is broadly considered as a valid concept for modeling water flow in roots, although advances in plant physiology and improvement in experimental methods for observing the rhizosphere emphasize that soil and plant resistances are all potentially variable in time and space (Hose et al., 2000; Carminati and Vetterlein, 2012). Therefore, the quantification of resistances to water flow along the pathway between the soil and the atmosphere, in particular between the rhizosphere, the cortex, and the xylem and between the leaves and the atmosphere through the stomata is still the subject of extensive research. The discussion arises from the fact that direct measurements of these local resistances are still hardly achievable today or their determination is still prone to large uncertainty. In addition, as already pointed by Hunt et al. (1991), differences in the definition of hydraulic resistances and their associated units adds confusion in the discussion. The uncertainty concerning the magnitude and location of these resistances led in the past to simplifying, sometimes simplistic modeling approaches for root water uptake (RWU).

Typically, root water uptake is accounted for in the Richards equation with a sink–source term S [$L^3\ L^{-3}\ T^{-1}$]:

$$\frac{\partial\theta}{\partial t} = \nabla\cdot\left[\mathbf{K}(h)\nabla h\right] + \frac{\partial K(h)}{\partial z} + S \qquad [1]$$

where θ denotes the volumetric soil water content [$L^3\ L^{-3}$], t the time [T], z the vertical coordinate [L], h the soil water pressure head [L], $\mathbf{K}(h)$ the soil hydraulic conductivity tensor [$L\ T^{-1}$]. In the right-hand part of Eq. [1], the two first terms describe the water flow redistribution between layers or soil locations, while the third one describes the water uptake by plant roots ($S < 0$) or root exudation ($S > 0$). This term may differ between authors in terms of dimensionality (Vrugt et al., 2001) but also in terms of complexity and degree of mechanical/biological principles explicitly accounted for (de Willigen et al., 2012). Two main approaches exist today to predict this sink term. On the one hand, physically based models may explicitly consider the three-dimensional distribution of the root system together with a distribution of the system conductances. These models require numerous input parameters that can in principle be measured. However, they are demanding in terms of computation power, memory. and parameters. On the other hand, effective models exist that represent the uptake behavior at the plant scale through macroscopic parameters. These functions are usually simple, with few parameters and easy to compute, but some of their parameters need calibration because they do not have physical or biological meaning. In this macroscopic approach, the sink term S is typically composed of four different (usually multiplicative) variables or terms that affect the magnitude and spatiotemporal dynamics of root water uptake, such as (i) the root hydraulic resistance distribution often represented by the root length density distribution, (ii) the soil hydraulic resistance distribution often represented by a function of the bulk soil matric and/or osmotic potential, (iii) a stress function describing the reaction of plants to an excessive climatic demand of water represented by transpiration rate dependent threshold values of water potentials, and (iv) a compensation function (Jarvis, 1989) representing the impact of the water potential distribution inside plant xylem vessels on the distribution of water uptake from the soil profile. In three-dimensional detailed models, the two first variables are explicitly considered by accounting for the distribution of the root architecture and the root and soil hydraulic properties (potentially changing in time, also). The third variable (stress) is defined by a function of the water potential in the leaf. The fourth variable (compensation) arises from the solution of the flow equations that are coupled between the root and the soil systems.

As both approaches aim at simulating the same processes, there should be more than conceptual links between three-dimensional explicit models and one-dimensional macroscopic parameters. Here we demonstrate that a three-dimensional mechanistic model that contains our current knowledge on biophysics can help understand the shape of the four variables of macroscopic model sink terms and define new functions or parameters that are closer to the state of knowledge on plant physiology and biophysics. By comparing three-dimensional simulations to one-dimensional functions, we outline underlying assumptions and define limitations of some macroscopic approaches and propose new ways of defining these four terms of the macroscopic sink-term function.

Materials and Methods

Three-Dimensional Water Flow Model in Soil and Roots

In the next sections, we analyze the role and impact of these four components of the sink term on root water uptake by embedding them in a three-dimensional modeling approach. Our assumption is that the three-dimensional root architecture and its hydraulic properties play a central role in the simulation and prediction of root water uptake, in particular because of the nonlinearity of the processes and the complexity of the spatial distribution of the properties. We will therefore use three-dimensional architectural models to simulate plant transpiration and water uptake.

Upscaling of the Ohm Analogy for a Complex Root System

Gradmann (1928) was probably the first to recognize "the existence of systematic gradients of potential in the plant and atmosphere domains with a continuity of potentials at the interfaces" (Philip, 1966) and propose the Ohm analogy for water flow in roots and between soil and roots (Van den Honert, 1948; Cowan, 1965). Since then, several models of root water uptake were developed on this basis (e.g., Nimah and Hanks, 1973; Molz, 1976; Landsberg and Fowkes, 1978; Alm et al., 1992; Lhomme,1998; Janott et al., 2011; Guswa, 2012; Volpe et al., 2013). Doussan et al. (1998a,b, 2006) proposed a numerical solution in which the three-dimensional water flow equations through the full root system are solved by Ohm analog laws in a tree-like structure in which lateral and longitudinal resistances (constant or variable) represent radial and axial hydraulic plant root resistances. Recently, Couvreur et al. (2012) reformulated this approach to extract a simple macroscopic stress function and a sink term distribution function, which predict the distribution of the water uptake and the reduction of the transpiration under water stress. Their model is thus a solution of the Ohm flow analogy for a full three-dimensional architecture into a non-uniform soil water heads (i.e., the total water potential in weight basis) field using three macroscopic parameters (SUF, K_{rs}, and K_{comp}) to represent plant behavior in a macroscopic model. The first equation relates the transpiration rate of the plant, T_{act} [$L^3\ T^{-1}$], to the equivalent conductance of the root system, K_{rs} [$L^3\ P^{-1}\ T^{-1}$], and the difference between the effective water head that is felt by the root system, $\tilde{H}_s$ [L], and the water head at the root collar H_{collar} [L]:

$$T_{act} = K_{rs}\gamma(\tilde{H}_s - H_{collar}) \quad [2]$$

where γ is the water specific weight [P L^{-1}]. The soil water head sensed by the plant is related to the local water head at the soil–root interface of the different root segments, $H_{sr,j}$ [L], as:

$$\tilde{H}_s = \sum_j \mathrm{SUF}_j H_{sr,j} \quad [3]$$

where SUF_j [-] is the standard uptake fraction, or the relative water uptake from root segment j for a homogeneous soil water head, which is directly obtained from solving the Doussan equations. The **SUF** vector contains weighting factors used to average the heterogeneous root–soil interface water head distribution into an effective unique water head felt by the plant and accounts for the spatial distribution of root hydraulic resistances.

The second equation describes the local uptake rate of root segment i [$L^3\ T^{-1}$]:

$$Q_{r,i} = T_{act}\mathrm{SUF}_i + K_{comp}\gamma(H_{sr,i} - \tilde{H}_s)\mathrm{SUF}_i \quad [4]$$

where K_{comp} is the compensatory RWU conductance [$L^3\ P^{-1}\ T^{-1}$]. The SUF, K_{comp}, and K_{rs} are root hydraulic architecture parameters that represent properties of the root system but that are independent of the soil water heads or transpiration rates. Equations [3] and [4] are the outcome of applying Ohm law analogy in a complex root system with nonuniform resistance and boundary conditions. It should be noted that water head at the soil–root interface is used in this model and not the bulk soil water head. For a given distribution of water heads at soil–root interfaces, the compensatory root water uptake (second term at the right hand side of Eq. [4]) does neither depend on the transpiration rate nor on soil hydraulic properties.

Explicit Three-Dimensional Model Accounting for Variable Soil and Root Resistance Distribution

Similarly to the Doussan or Couvreur equations, which solve the flow equation in the three-dimensional root system, the Richards equation can be solved numerically to represent the three-dimensional water flow distribution in soils. In this study, we use the model R-SWMS (Root–Soil Water Movement and Solute transport platform), which simulates water flow in the coupled soil–root system based on water head differences and gradients. This model solves the Richards equation numerically for soil water flow (Simunek et al., 1995) and the Doussan equations for water flow within the root xylem system and between soil–root interface and xylem vessels (Doussan et al., 1998a) in a three-dimensional root system, and couples both systems through sink terms. A description of this model is given in Javaux et al. (2008). The main assumptions in this model are that (i) the water flow is driven by gradients of water head in soil and roots, (ii) that soil and root hydraulic properties can be described in an effective way respectively at the soil voxel and the root segment scales (with a typical length scale of ~1 cm), and (iii) that the uptake can be described by a sink term at the voxel scale, inside which soil water head heterogeneity can be neglected. It can therefore be used to simulate plant behavior under changing boundary conditions and with root parameters which depend on plant species, age or environmental factors. Scenarios and parameters used in this paper are summarized in Table 1. In the following, we assume that this model can represent the result of the Richards and Doussan equations coupled through a sink term and can be used a benchmark for macroscopic models.

Root Resistance and Plant Hydraulics

State of the Art: Description of Flow Processes within the Root System

In soil, the hydraulic conductivity tensor $\mathbf{K}(h)$ (in L T^{-1}) relates the water flow to the gradient of the hydraulic head and it describes the ability of the porous medium to conduct water in three dimensions. In plant physiology, different resistances/conductances can be defined according to the tissue that is crossed or the scale of interest. In the following, units and definition of resistance, conductance, conductivity, and resistivity will be given in agreement with the definitions of Hunt et al. (1991) given in the appendix.

When entering the root, water must flow through the complex anatomical structure of root cortex to reach the xylem vessels, through which water is conducted to the shoot and the leaves

Table 1. Parameters and boundary conditions of the simulations.

Figures	Scenarios	Root system	Root hydraulic properties	System geometry	Soil properties	Model	Duration and plant collar condition†	Soil initial conditions
Impact of root hydraulic architecture								
1	1a	Fibrous, generated with RootTyp. See Schröder et al. (2012) for details.	Uniform $K_r = 5 \times 10^{-8}$ s^{-1} $K_x = 0.2 \times 10^{-6}$ cm^3 s^{-1}	Root system in a pot	No explicit soil properties	Doussan	Instantaneous H_{collar} = −2500 cm	Linear distributions of water heads at soil root surface
	1b	Taproot, generated by RootTyp. See Schröder et al. (2012) for details.	Uniform $K_r = 1.61 \times 10^{-9}$ s^{-1} $K_x = 5 \times 10^{-5}$ cm^3 s^{-1}	Root system in a pot	No explicit soil properties	Doussan	Instantaneous H_{collar} = −2500 cm	Linear distributions of water heads at soil root surface
Maize water extraction under field conditions								
3–4–5–6	2a	Maize root system generated by RootTyp. See Couvreur et al. (2012) for details.	Function of root age and type (Doussan et al., 1998b)	*XY* Periodic soil and plant BC. ($dx = dy = dz$ = 1.5 cm)	Silt loam (from Carsel and Parrish 1988), without air gap	RSWMS	26 d variable day–night transpiration	Static equilibrium H_s = −300 cm
	2b	Maize root system generated by RootTyp. See Couvreur et al. (2012).	Function of root age and type (Doussan et al., 1998b)	*XY* Periodic soil and plant BC. ($dx = dy = dz$ = 1.5 cm)	Silt loam (from Carsel and Parrish 1988), with air gaps	RSWMS	26 d variable day–night transpiration	Static equilibrium H_s = −300 cm

† H_{collar}, water head at the root collar; H_s, soil water head.

(Lobet et al., 2013). The composite model proposed by Steudle and Peterson (1998) states that water can either flow through the porous cell walls (apoplastic pathway) or through cells (symplastic and cell-to-cell pathways), thereby crossing cell membranes thanks to aquaporins (Maurel and Chrispeels, 2001). Yet, due to presence of impermeable cell walls in the casparian band, the existence of a purely apoplastic way is questionable (Knipfer and Fricke, 2010). In such case, osmotic potential differences across cell membranes may play an important role in driving water flow. The water flux through a cell membrane, $q_{w,membrane}$, [L^3 L^{-2} T^{-1}] is expressed by the following membrane equation:

$$q_{w,\,membrane} = L_{p,cell}\left[\psi_{p,ext} - \psi_{p,\,in} + \sigma\left(\psi_{o,ext} - \psi_{o,in}\right)\right] \quad [5]$$

where $L_{p,cell}$ is the cell membrane hydraulic conductivity [L T^{-1} P^{-1}], $\psi_{p,ext} - \psi_{p,in}$ and $\psi_{o,ext} - \psi_{o,in}$ are, respectively, the differences of pressure and osmotic potential [P] across cell membrane, and σ is the membrane reflection coefficient. It has been shown that cell membrane conductivity may be actively changed by plants (Javot and Maurel, 2002), triggered by the expression and specific activation of the aquaporins.

By scaling up this equation to the complete radial cortex, the radial water flow is defined as (Fiscus, 1977):

$$J_{w,\,r} = K_r s_{int} \gamma\left[h_{p,soil} - h_{p,x} + \sigma\left(h_{o,soil} - h_{o,x}\right)\right] \quad [6]$$

where $h_{p,soil}$ and $h_{p,x}$ are the pressure heads respectively at the root surface and in the xylem [L], $h_{o,soil}$ and $h_{o,x}$ are the osmotic heads respectively at the root surface and in the xylem [L], K_r is the root radial conductance [L P^{-1} T^{-1}], σ the membrane reflection coefficient, and s_{int} is the root–soil interface area [L^2]. The effective radial conductance is influenced by the organization of the cells (i.e., size, locations) and by the spatial distribution of the cell wall and cell membrane hydraulic resistances. The effective resistance therefore differs between plant species, root developmental stages, and function and type (Pages et al., 2004; Pierret et al., 2007), or changes in root anatomy and cell permeability (Maurel et al., 2010). The latter was shown to be a rather quick mechanism in a time span of about several minutes to a few hours (Hachez et al., 2012).

The xylem flow can be defined with a formula similar to the Darcy equation. Xylem conductance is much higher than that of the radial tissues, which led several authors to neglecting xylem impedance to water flow (Steudle and Peterson, 1998). Yet, this is not always correct since the length of the flow path through the xylem route is much longer than the radial flow distance. Furthermore, xylem conductivity may decrease considerably due to cavitation (Sperry et al., 2002, 2005). Xylem conductance and its change over time depends on a longer time scale on root development and order, which define the length, connectivity, and diameter of xylem vessels whereas on the short time scale, it depends on the onset of embolism. At the plant scale, root architecture controls the distribution of radial and xylem hydraulic properties and how root segments are connected (Draye et al., 2010). The root architecture is the result of both plant genetic endogenous programs (regulating growth and organogenesis) and the reaction to environmental stimuli (Hodge et al., 2009). In that regard we can expect that plant architecture will partly reflect the environmental conditions to which it is adapted (Pierret et al., 2007). By defining the soil locations to which roots have access, root architecture has a non-negligible impact on current uptake. By imposing the location of root tips (and meristems), root architecture also defines the ability of plants to respond to heterogeneous water distribution by growing new roots from certain specific locations.

The concept of root hydraulic architecture (Couvreur et al., 2012) integrates at the same time the three-dimensional root architecture (the root system topology) and the specific spatial distribution of root hydraulic properties (axial and radial conductances) of a plant root system.

Current Macroscopic Approaches for Root Resistance and Open Questions

The easiest way of considering root resistances in a macroscopic model that does not consider individual roots is to assume that roots all have constant and uniform radial conductance and that the xylem resistance is negligible. In that case, the water potential is uniform inside the root system so that the conductance for flow into the root system at a certain location is proportional to the local root density as in most macroscopic models (Nimah and Hanks, 1973). Note that, when the root surface (Volpe et al., 2013) or weight density profiles are used instead of the root length density (RLD), it is implicitly assumed that the radial conductance is respectively a one- or second-order function of the root diameter.

The use of a one-dimensional root profile instead of the three-dimensional root hydraulic architecture in macroscopic models means that root hydraulic architecture is assumed not to be relevant for water uptake, that is, only the amount of roots matters but not how root segments are connected or what is the distribution of their hydraulic properties. This implies that the temporal evolution of the plant root conductance is only related to root growth, thereby neglecting that new root segments may have different hydraulic properties than the rest of the system. Yet, two plants may have similar RLD profiles but very different root architecture (see, e.g., Schröder et al., 2012), and thus access to water. When using one-dimensional RLD profiles, this also implies that the horizontal variability in root and soil properties is negligible. However, the location of roots in the horizontal cross section (and their hydraulic properties) may be of importance for the uptake.

Discrepancies in the water content profile evolution between predictions with RLD models and experiments are sometimes explained by the fact that not all roots are efficient. Faria et al. (2009) for instance estimated the root system efficiency at about 5%. This concept of root activity or root efficiency is usually defined as the actual uptake of nutrients or water by roots as compared to root length density (Lehmann, 2003). Active roots are the roots that actually take up water (or nutrients). From Eq. [6], it is obvious that the activity of a root segment for water uptake is not an intrinsic root property but will depend on the hydraulic head gradient between soil–root interface and xylem and on root segment conductance, which are terms not considered in macroscopic models.

Note that even without changes in root intrinsic conductance, root water uptake (or activity) will be highly variable with time and space, as hydraulic heads will vary with transpiration intensity and soil hydraulic status. Root activity is therefore not only linked to static root conductance properties. It is intrinsically linked to the definition of water uptake and of compensation, which will be further discussed in that section.

Analysis and Perspectives: Impact of Root Architecture

The definition of an effective plant hydraulic conductance K_{rs} in Eq. [2] accounts for the distribution of local radial and axial conductances and for the plant root architecture. Basically, K_{rs} can be calculated using the Thevenin theorem (Thévenin, 1883), providing the equivalent conductance of conductor networks. In our case, the network is the root architecture, and local conductances are defined by local xylem and cortex hydraulic properties. In case xylem resistance is negligible, K_{rs} is equal to the sum of local radial conductances, as in an electrical circuit with resistances in parallel, and is only sensitive to root hydraulic properties. In case xylem resistance cannot be neglected, the shape of the network needs to be taken into account, and K_{rs} becomes sensitive to root system architecture too. The equivalent root system conductance K_{rs} links the actual transpiration T_{act} [$L^3\ T^{-1}$] to the difference between the water head at the plant collar (or leaf when the complete plant is accounted for) and soil water head sensed by the root system.

The impact of root architecture on the soil water uptake profile is investigated by using Eq. [2] to simulate the actual transpiration of a tap-rooted and a fibrous root system having the same K_{rs} (Fig. 1a and 1b). To obtain the same K_{rs}, the xylem conductance of the tap root was defined to be high while we used a lower xylem conductance for the fibrous system (Table 1). We assumed a horizontally uniform distribution of the soil water heads at soil–root interfaces $H_{sr,j}$. We tested a set of vertical gradients of $H_{sr}(z)$ ranging between -40 cm cm^{-1} (drier at the soil surface) to 40 cm cm^{-1} (wetter at the soil surface), all of them having the same profile average water head equal to -1000 cm. The collar water head was set at -2500 cm.

Although having a similar root length density profile and K_{rs}, these two root systems differ in the SUF distribution (Fig. 1c), which was estimated by solving the root water flow equation (Doussan et al., 1998a). Note that the RLD and SUF profiles of the fibrous system differ due to its limiting xylem resistance. Figure 1 shows that the

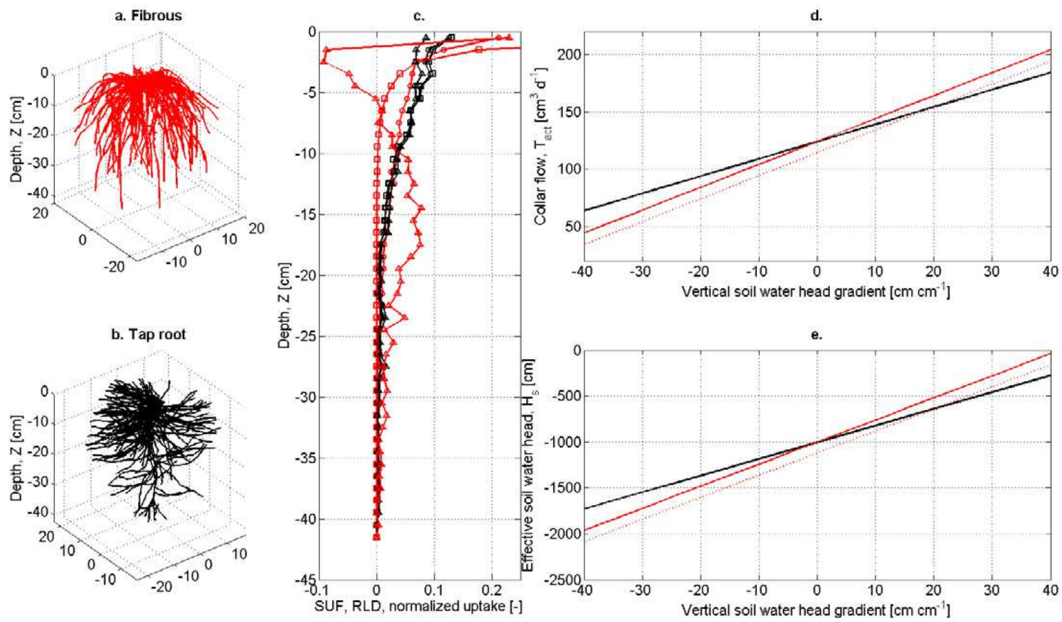


Fig. 1. Comparison between a fibrous (red lines) vs. taproot root system (black lines): (a) fibrous root system, (b) tap root system, (c) vertical distribution of the root length density (RLD, continuous lines with circles) and standard uptake fraction (SUF, lines with squares) and normalized uptake distribution under a vertical pressure head gradient of 40 cm cm^{-1} (lines with triangles), (d) actual root collar flow without (solid lines) and with radial gradients (dashed line), (e) effective root-sensed soil water potential without (solid lines) and with radial gradients (dashed line).

pressure heads at the root collar are more sensitive to the water head in the upper soil layer than in the deeper soil layer; that is, the root-sensed effective water head is predominantly controlled by depths where there are more roots, because the corresponding SUF is higher (Fig. 1e). Thanks to its more uniform SUF, the tap-root system is less sensitive to vertical pressure head gradients than the fibrous root system (Fig. 1e). When the upper soil is drier, the tap-root system transpires more than the fibrous system (up to 25% of transpiration for vertical gradient equal to −40, Fig. 1d) and this is the opposite when the gradient is inversed. Looking at vertical profiles (Fig. 1c), the fibrous system SUF is very different from its RLD and of the normalized uptake profile under a gradient of −40 cm cm^{-1} (with hydraulic lift). On the opposite, the taproot system has a similar RLD, SUF, and normalized uptake. When radial gradients of hydraulic head are added (depletion in the center, higher water head in the outer ring), this impact is even bigger for the fibrous root (Fig. 1d-e, red lines), while no effect is observed for the tap root (Fig. 1d–1e, the black dashed line is overlapped by the continuous black line) since most of its lateral roots cross the soil radially and sense the wetter lateral zones. With this simple example, we showed that the hydraulic architecture summarized in the SUF may thus play a role when water heads become heterogeneous (vertically or/and horizontally) and in case xylem resistance cannot be neglected. This is typically the case for dry periods in maize fields, when a dry soil zone develops below the plant, while the outer zone is wetter (Beff et al., 2013), and xylem cavitation may occur, making root axial resistances even more limiting.

It is expected that the impact of root hydraulic architecture will be even larger under real conditions, when the drop in soil hydraulic conductivity for dry soil may limit root water uptake. Using a model similar to RSWMS, Schneider et al. (2010) showed that under drying conditions the difference in RWU between root architectures was increasing with time. With time, the water head profile will become more heterogeneous and, in line with simulation results shown in Fig. 1, amplify root architecture differences.

To summarize, root length density distributions may be used to describe the distribution of root water uptake but only in case of low xylem hydraulic resistance and homogeneous root radial resistances, as for the example, with the taproot system in Fig. 1c. In other cases, spatial variations in RWU will lead to an increase of horizontal and vertical variability of matric head over time, and it is expected that sink routines using the RLD will fail to predict accurately root water uptake profiles and stress onset. However, the use of the hydraulic architecture (SUF instead of RLD) could easily be implemented in macroscopic models and would improve their reliability.

Soil Resistance

Current Modeling Approaches and Challenges

The local water uptake is driven by the water head difference between soil–root interface and xylem pressure head, as shown in Eq. [6]. However, the local soil–root interface water head is quite complex and difficult to measure (we measure bulk properties and variables), and in addition the pressure head at the soil–root interface is also dependent on the soil conductivity around roots. Thus, models have been developed to account for the impact of soil properties on root water uptake. Molz (1981), among others, reviewed several sink term functions that explicitly account for soil water flow resistance, some of them being still in use today. Most of these functions account for soil resistance through the soil conductivity or diffusivity functions and consider flow that is driven by pressure head or matric flux potential gradients around plant roots (Gardner, 1960; de Jong van Lier et al., 2008). Other functions exist where soil resistance is simply derived from the soil conductivity, which is calculated using the bulk soil water content. (e.g., Katul et al., 1997).

The water flux density q_{int} [L T^{-1}] reaching the soil–root interface around a cylindrical root of radius r_{int} in a circular domain of radius r_{bulk} at which the outer flux is q_{bulk} can be expressed as (Schröder et al., 2009):

$$q_{\text{int}} = \frac{B}{r_{\text{int}}}\int_{h_{\text{int}}}^{h_{\text{bulk}}} K(h)\mathrm{d}h + B\chi_1 + \chi_2 \qquad [7]$$

where

$$B = \frac{2\left(1-\rho^2\right)}{-2\rho^2\left(\ln\rho - \frac{1}{2}\right) - 1}, \quad \rho = \frac{r_{\text{bulk}}}{r_{\text{int}}}, \quad \chi_1 = q_{\text{bulk}}\rho\ln\frac{1}{\rho}, \quad \text{and } \chi_2 = q_{\text{bulk}}\rho.$$

When the outer flux is zero, the second and third term of Eq. [7] cancel. Equation [7] can be used with two different r_{bulk}. Some authors consider r_{bulk} as being one-half the average distance between different plants, and thus q_{int} represents the total plant uptake—this is the big root approach. Others consider that r_{bulk} represent one-half the average distance between two root segments that can be deduced from the RLD profiles. The actual transpiration is then the sum of Eq. [7] for all root segments—this is the approach de Jong van Lier et al. (2008) used for instance.

We investigated with Eq. [7] how the soil water head at the interface h_{int} was dependent to the RLD, through the definition of $r_{\text{bulk}} = \sqrt{1/\pi \text{RLD}}$ and $q_{\text{int}} = S/2\pi r_{\text{int}}\text{RLD}$ for different soil types of the database of Carsel and Parrish (1988). The following parameters were used: root radius r_{int} = 0.05 cm, bulk water head h_{bulk} = −600 cm, and sink S of 0.02 cm^3 cm^{-3} d^{-1} (corresponding to a transpiration rate of 6 mm d^{-1} for a rooting depth of 0.3 m).

The values of the water head at the soil–root interface compared to the bulk pressure head (= −600 cm) represent the head loss due to resistance to flow in the soil towards the root surfaces for different typical soil types and RLD. When root density decreases the flow per root q_{int} increases and hydraulic gradients develop in the soil. It is observed that important gradients may exist between bulk soil and soil–root interfaces, in particular in coarse soils and with low active root length density values (which are commonly observed in real soils). In such cases, the fluxes in the root system and the corresponding head losses may be so large that the flux cannot be supported by the system due to too negative pressure heads at the soil–root interface. If the transpiration rate is controlled by stomatal closure and if the stomata control the pressure at the root collar so that it does not fall below a certain threshold limit, the corresponding pressure head at the soil–root interface can be calculated for a given radial flux density $q_{\text{int}} = J_{\text{w,r}}/s_{\text{int}}$ using Eq. [6]. For the sink term defined above, using a K_{r} = 1.728 10^{-4} d^{-1}, and assuming that the xylem conductance is very large and that there is no osmotic gradient, the pressure head at the soil–root interface can be calculated for the different root length densities and is given

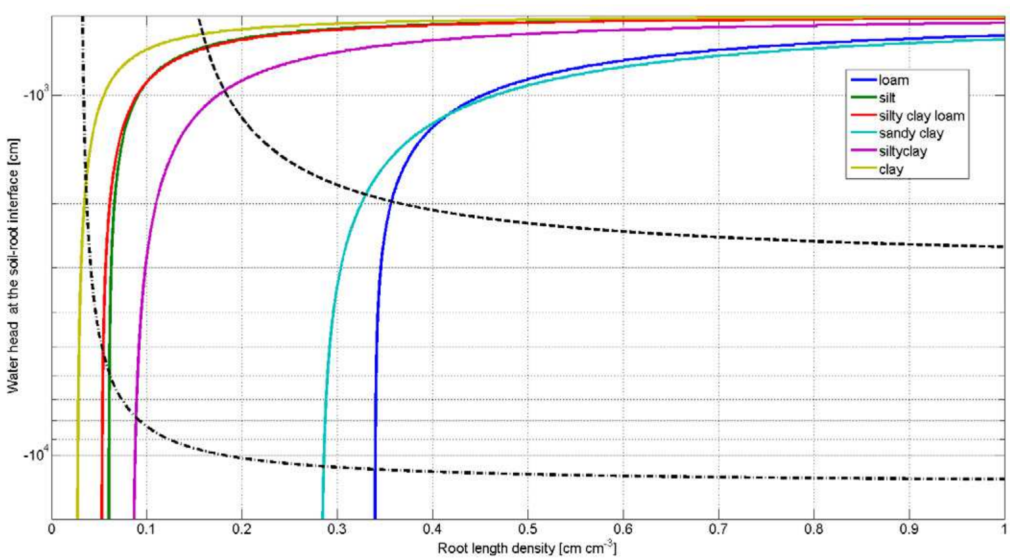


Fig. 2. Soil water head at the soil-root interface in function of the root length density estimated from Eq. [7] for several typical soil types (color lines) for a sink of 0.02 cm^3 cm^{-3} d^{-1} and a bulk water head of −600 cm. Black lines show Eq. [6] in which root pressure head is −12000 cm (dotted-dashed line) and −3000 cm (dashed line).

for a collar water head $h_{\mathrm{collar}} = -12{,}000$ cm (dashed-dotted line) and for $h_{\mathrm{collar}} = -3000$ cm (dashed line) in Fig. 2. These curves represent the minimal pressure heads at the soil–root interface to drive the flow in the plant towards the collar for the given collar pressure heads. From the intersects of the curves in Fig. 2, the minimal root length density and the corresponding soil–root interface pressure head that are required to avoid a reduction in water uptake from a soil with a bulk soil pressure head of −600 cm, and for a sink term of $-0.02\ \mathrm{d}^{-1}$ and a minimal pressure head at the root collar of −12000 or −3000 cm can be derived. This plot indicates that whether or not water uptake will be reduced for these bulk soil conditions depends on the soil type, root length density, and the minimal allowed pressure head at the root collar. Therefore, besides the bulk soil water head, which is used in macroscopic models to represent the effect of soil hydraulic resistance on local root water uptake, also soil type, root length density, and sink term or transpiration rate influence the effect of soil resistance on root water uptake. It should be noted that transpiration rate is already included in stress functions that describe local root water uptake reduction as a function of the local bulk soil pressure head, whereas the effect of root length density, conductance of the root system, and critical pressure head in the leaves may be represented by a plant or crop type specific parameterization of the stress function.

Another question is whether soil properties around roots differ from the bulk soil. Indeed, the impact of the rhizosphere, which refers to the soil zone that is intimately linked to the root, on root water uptake has been debated for a long time (So et al., 1976; Herkelrath et al., 1977). Today it is evident that it plays an important role for water and solute uptake under certain circumstances. Shrinking–swelling of root and soil (Carminati et al., 2009), soil compaction due to root growth (Aravena et al., 2011), mucilage exuded by root caps, interaction of mucilage with soil particles (Moradi et al., 2012), mucilage shrinking–swelling, and change of wettability (Carminati, 2013) are rhizosphere processes that will eventually impact soil properties around roots, and thus, water (and solute) uptake (Hinsinger et al., 2005; Carminati and Vetterlein, 2012).

Perspectives: Impact of Rhizosphere Properties on RWU

A simulation was performed with R-SWMS, where the impact of the development of air gaps around roots when soil dries out is implicitly accounted for. We considered that root–soil contact was not lost abruptly because root hairs and the irregular shape of roots and soil pores will probably keep contact for a while before soil and root disconnect. We assumed that the loss of contact was proportional to the local water head and therefore linearly related to a decrease of the radial conductance by a factor f set to 1 when soil water head is −500 cm and 0 when $H = -15{,}000$ cm.

Figure 3 shows that air gaps tend to decrease the collar water head during the day and increase it during night. During the day, when roots lose contact, the total soil–root conductance decreases and xylem head must decrease as well to sustain the same transpiration flux. During night, though, air gaps help plants to be insensitive to the driest soil zones so that the root collar water head remains higher. When no air gaps appear, at the beginning of the scenario, the effective soil water head felt by the plant decreases during the day, due to soil drying. While at the end of the scenario, the opposite dynamics can be observed: during the day, due to soil drying, air gaps form, and the soil water head felt by the plant increases, since it does not feel the drier zones anymore. Conversely, during the night, due to soil water redistribution, soil and roots swell back, and the plant feels the low soil water head again.

The impact is also visible on transpiration flux. The root system with air gaps starts suffering from stress first, and every day a bit earlier than the reference root. In total, the cumulative transpiration volume is lower for the root with air gaps, but in the long term, this could be a strategy to conserve water and to avoid too low water heads in the plant.

Stomatal Regulation and Transpiration Reduction

State of the Art and Modeling Approaches

Although it is known that stomatal closure is sensitive to multiple environmental influences like leave water status, air humidity, and temperature, there is no consensus among biologists on how environmental stimuli are sensed and transduced into the plant and eventually trigger stomatal response (Damour et al., 2010). The response to soil water deficit has been shown to be mediated directly through hydraulic signals transduced in the shoot (e.g., Comstock and Mencuccini, 1998) and indirectly by root-based chemical messages transported by the xylem stream (e.g., Dodd et al., 2010). Irrigation techniques like the partial root zone drying or deficit irrigation take advantage of the latter process. Whatever the transmission mode, it is expected that drier soils with lower water potential will induce stomatal closure and generate drought stress. Tardieu (1996) discriminated between two types of plant stomatal regulation: isohydric and anisohydric. Under water stress, the first group of plants (e.g., maize, *Zea mays* L.) controls their stomatal conductance to maintain leaf water status almost at a constant level, regardless of soil water status. On the opposite, the second group (e.g., sunflower, *Helianthus annuus* L.) does not regulate stomatal conductances in function of the leaf potential so that leaf water potential fluctuates proportionally to the soil water potential under a given atmospheric demand. In between these two extreme behaviors, stomatal conductance may be either a continuous function of leaf water potential or of other signals (Tardieu and Davies, 1993).

In soil hydrology, the regulation of stomatal conductance is linked to the soil water head, which is used as a proxy for the leaf water potential. Most of the approaches define a bulk water head or average water content below which plant stress starts and affects

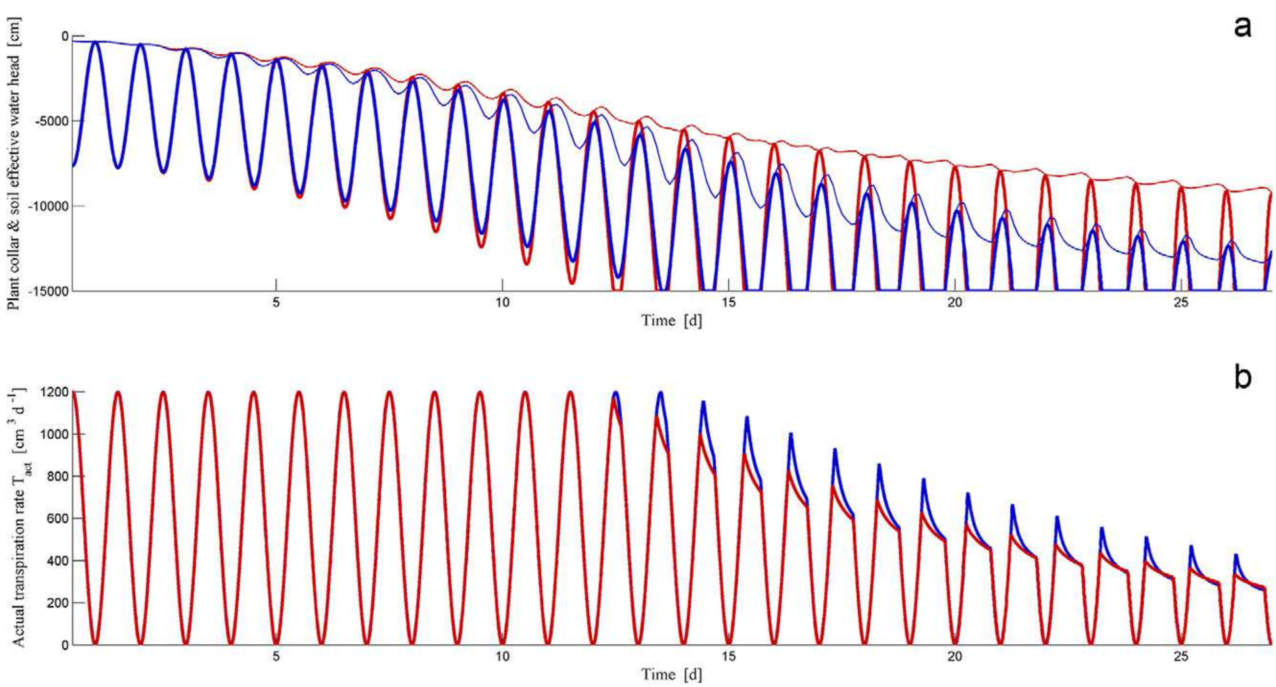


Fig. 3. Impact of air gap on root collar flow dynamics. Blue: reference scenario (2a in Table 1); Red: scenario with air gaps (2b in Table 1). (a) Plant collar water head (thick lines) and soil effective water head felt by the plant (thin lines), (b) actual instantaneous transpiration rate.

transpiration flow. The Feddes stress function is widely used for relating the decrease of transpiration to the bulk soil water head with a piecewise linear function (Feddes et al., 1976). In ecohydrological models, stress functions typically relate a decrease of the transpiration to the soil moisture, which is, through linear or nonlinear soil water retention curve, directly related to the soil water head (Porporato et al., 2002).

These approaches rely on several assumptions. First, by making the transpiration reduction a direct function of the soil water status and transpiration rate only, the impact of plant hydraulic architecture can only be considered through effective parameterization. Second, they usually assume a unique function for a given plant species independent of the plant status, which implies that they cannot account for changes or adaptations of root resistances and root architecture over time as a response to environmental conditions. Third, they assume implicitly that the water head in the soil is uniform horizontally.

New Perspectives: Simulation Results

We can use Eq. [2] to link actual transpiration, soil–root interface water head and leaf water head under stress conditions. Under isohydric control, we can assume that the collar water head of Eq. [2] is kept constant by a continuous adjustment of the stomatal conductance by plants ($H_{\mathrm{collar}} = H_{\mathrm{lim}}$, a limiting water head level). This may be the case when the stomatal conductance is a strongly nonlinear function of the leaf water pressure and decreases rapidly when a certain threshold water head is reached (Tuzet et al., 2003). In such case, a water stress function writes

$$\frac{T_{\mathrm{act}}}{T_{\mathrm{pot}}} = \frac{K_{\mathrm{rs}}}{T_{\mathrm{pot}}}\gamma\left(\tilde{H}_{\mathrm{s}} - H_{\mathrm{lim}}\right) \qquad [8]$$

This equation shows a linear relation between $T_{\mathrm{act}}/T_{\mathrm{pot}}$ and plant-sensed soil water head. The slope of this relation is governed by the root K_{rs}, which summarizes the hydraulic architecture of the root systems, and the potential transpiration T_{pot}. For a given T_{pot}, this relationship may become nonlinear due to a change/adaptation of the plant equivalent conductance K_{rs} with $\tilde{H}_{\mathrm{s}}$.

The same scenarios as in the previous section are used for illustration of the stress (Table 1, Scenarios 2a and b). In Fig. 4, we plotted the actual transpiration versus the plant-felt water head for the scenario with (left) and without (right) air gaps at soil–root interfaces.

It is observed that day–night cycles of plant transpiration when soil is drying define an envelope of T_{act}–H_{s} combinations. For Scenario 2a, the limit of this envelope can be predicted by Eq. [2] with slope K_{rs} and intercept the limiting collar (or leaf) water head. In Scenario 2b, the appearance of air gaps at soil–root interfaces generates a decrease in soil–root interface conductances, which induces a decrease of K_{rs} with decreasing soil water head, making the limit of the stress envelope nonlinear. It is interesting to note that even if the air gaps generate an increase of the soil water head felt by roots as compared to the reference scenario (due to a different SUF in Eq. [3]), the plant begins to suffer from water stress earlier in time, and at higher sensed-water head, due to its lower radial conductances, which generate more tension in xylem (see Fig. 3).

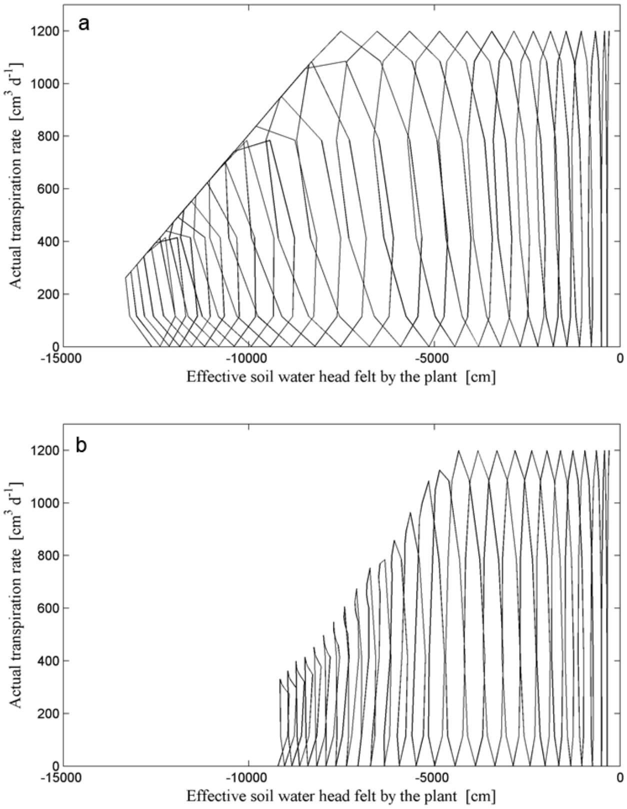


Fig. 4. Actual transpiration trajectories obtained from simulations with RSWMS for (a) maize reference scenario and (b) for the scenario with air gaps. See Table 1 for details on Scenarios 2a and b.

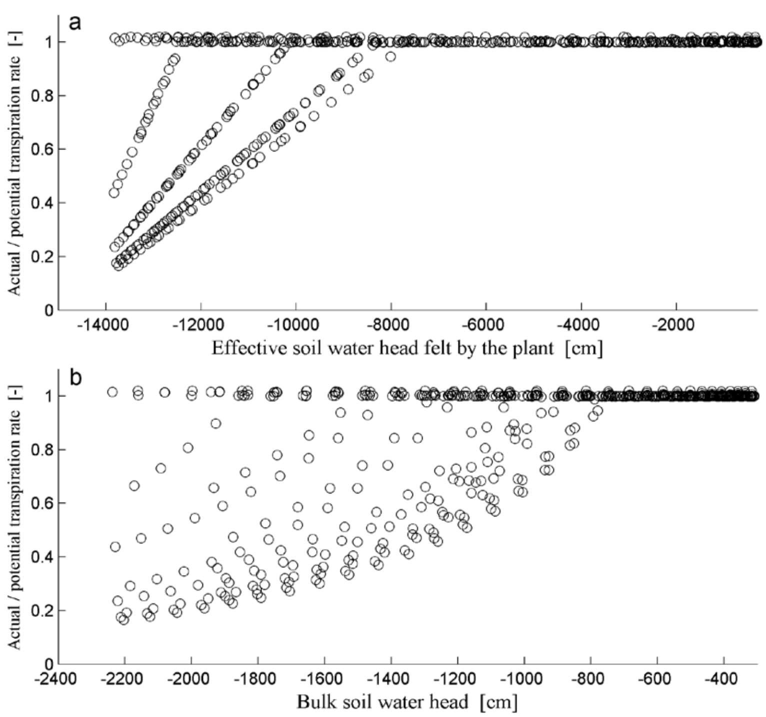


Fig. 5. Relation between transpiration rate reduction and soil water potential, obtained at discrete potential transpiration levels, from simulations with RSWMS for the maize reference scenario (2a in Table 1). Transpiration rate reduction vs. (a) effective soil water potential felt by the plant and vs. (b) bulk soil water potential.

In Fig. 4, it can be observed that the onset of stress starts at more negative soil water head (about −8000 cm) than what is typically considered for most crop stress functions (Taylor and Ashcroft, 1972; Wesseling et al., 1991). This can be explained by the fact that Eq. [10] uses the integration of soil water heads at soil–root interfaces, while Feddes considers the bulk soil water head or content (Feddes et al., 1976). The magnitude of that difference is in the same range as what we estimated in Fig. 2. In Fig. 5 the stress factor (Eq. [8]) is shown as a function of the soil–root interface water head (5a) or of the bulk soil water head (5b). The bulk soil water head was calculated as the water head corresponding to the mean water content of the root zone, on the soil water retention curve. It also corresponds to the water head that would be reached after equilibration of the water present in the root zone.

The impact of soil resistance is clearly visible, as there is almost a factor 10 in water head scales between both plots. In addition the nonlinear soil water head distribution around roots due to variable soil hydraulic conductivities generates nonlinearity and noise in the relation to bulk soil water head. Interestingly, it is observed that there are values of very negative soil pressure heads for which there is no reduction in T_{act}/T_{pot}: these correspond to very low transpiration rates for which, even when the soil is dry, there is no limitation by the soil conductivity.

This shows that the ratio of the actual to the head transpiration rate versus the soil water head is a function of the potential transpiration. Instead of using this ratio for the stress function, as in Feddes et al. (1978) for instance, the stress function should rather be defined as a maximal actual transpiration rate for a certain plant-sensed water head (as proposed by Jarvis, 2011). The slope of this T_{max} would be defined by the root–soil system conductance (Sperry et al., 2002); this would remove the dependency of the slope to T_{pot}. The stress function should also be defined at the plant scale and not as a local function, as mentioned by Jarvis (1989), since it depends on the stomatal conductance and on the xylem water head, but also on the distribution of soil water head at the soil root interface and root hydraulic architecture (Eq. [2] and [3]).

However, Eq. [2] does not consider explicitly the additional head loss between soil–root interface and bulk soil due to the decrease of $\mathbf{K}(h)$ with moisture content. To take into account the nonlinear relation between soil resistances and water heads, a solution of the Richards equation must be used, such as that based on the bulk matric flux potential as proposed by de Jong van Lier et al. (2008). When the stress function is defined as a function of the bulk soil pressure head (like in typical one-dimensional models), the nonlinearity of the soil hydraulic conductivity or soil resistance leads to a nonlinear dependence of the maximal possible transpiration rate on the bulk soil pressure head. Furthermore, this relation will depend on the soil hydraulic properties and the horizontal and vertical root distribution in the soil. Heterogeneity of the root length density across a horizontal soil surface will generate additional heterogeneity in soil water heads distribution that cannot be considered in analytical solutions of the radial water flow equation towards roots such as Eq. [7]. We speculate that lateral variations in root density and in root hydraulic properties and the corresponding spatial variation of water potentials will, due to the nonlinearity of the flow process, generate an earlier reduction of the transpiration rate when the soil dries out than for the case of uniform root properties and distributions. This may be an explanation for the fact that often lower active root length densities than actually observed ones are required to relate predictions by analytical equations to observations of root water uptake. Finally, since soil water head distributions in the root zone are also influenced by rainfall and irrigation distributions during the growing season, we speculate that relations between water uptake and bulk soil water head will also depend on these factors.

Compensation and Root Connectivity

Process Definition

The compensation mechanism is defined by several authors as a process by which plants "balance reduced water uptake from one part of the rhizosphere by increased uptake in another less-stressed region of the rooting zone" (Hopmans and Bristow, 2002). Yet, the definition of a stressed region is not very clear and is most probably a soil physics centered definition, since the soil is not stressed per se. In addition it is also questionable whether the plant can suffer from drought stress in a part of its root system, as usually stress is defined at the level of the plant (Tardieu, 1996). There is thus a need for a more clear definition.

We define compensation as a process by which root water uptake is spatially redistributed due to a nonuniform water head distribution at the soil–root interface (root hydraulic redistribution). This is a passive mechanism, whose driving force is the gradient of water head between soil–root interfaces of a plant, and controlled by the distribution of root hydraulic conductances, as predicted by the water flow equation in three-dimensional. In that regard, hydraulic lift, that is, the mechanism by which water moves from wet to dry soil regions via the root system, is due to compensation (Prieto et al., 2012; Jarvis, 2011). Compensation is thus independent of water stress and may occur even under relatively wet conditions, as soon as soil water head distribution is not uniform. Several authors suggest that active mechanisms could regulate or adapt plant conductances, which could enhance or impede further this process.

Current Modeling Approaches

In the initial Feddes model (Feddes et al., 1976, 1978), root water uptake is modeled as a local process. This means that uptake at a certain location is not influenced by the soil status or root properties at other locations. However, since the root system is a connected system, it is evident that water uptake by roots is a nonlocal process. This means that the uptake at one location depends also on the water status at other locations (Nimah and Hanks, 1973; Jarvis, 2011; de Jong van Lier et al., 2008). For instance, uptake affected by non-local soil status may be important when a part of the root zone has dried out (e.g., due to top soil drying) or a part has been wetted (e.g., due to partial root zone wetting in drip irrigation). An extreme case of non-local root water uptake is hydraulic lift, that is, the process by which water is exuded from the root due to inverted gradient potential. Concepts to address this non-locality have been implemented in models, but these concepts are not always based on any mechanistic description of the process (Li et al., 2001) or on the root architecture (Lhomme, 1998). Interestingly, Guswa (2012) recently proposed a semimechanistic equation accounting for compensation (see their Eq. [7]), which has similar shape than the second term of our Eq. [4] when divided by T_{pot}, but based on bulk one-dimensional variables and slightly different parameters.

A typical modeling approach that is proposed in soil hydrology to account for compensation in one-dimensional macroscopic models is to link it with the local reduction of water uptake when the local soil resistance, which is a function of the local soil water head, impedes water flow towards the roots (Šimůnek and Hopmans, 2009). A global weighted stress index can be defined as (Jarvis, 1989):

$$\omega(t)=\int_0^L \alpha(h)b(z)\,\mathrm{d}z \qquad [9]$$

where $\alpha(h)$ ($0 < \alpha < 1$) is a local flow impedance function characterizing the impact of soil limitation to water uptake, commonly represented by the Feddes stress function (Šimůnek and Hopmans, 2009), $b(z)$ ($0 < b < 1$) is the relative root length density [L^{-1}], L is the maximum rooting depth [L]. This index varies from 0 to 1. Jarvis (1989) defined a compensated sink term as

$$S_c(t)=S_p\frac{\alpha(h)}{\max\left[\omega(t),\omega_c\right]} \qquad [10]$$

where S_c [T^{-1}] is the sink term for compensated water uptake, $S_p = [b(z)T_{pot}]$ is the potential sink term [T^{-1}], and ω_c is the compensation factor.

Local water uptake is thus only reduced if transpiration reduction or stress occurs when the whole root zone experiences this local water head. For $\omega = 1$, the root water uptake profile (represented by the sink term distribution) is proportional to the root profile, meaning that the soil is not limiting water uptake in any parts of the rooting zone. When $\omega < 1$, soil is limiting the uptake at least at one depth and sink and root profiles differ. Compensation may then occur and increase the uptake from layers where the soil is wetter. Compensation may then potentially occur, following the value of a user-defined compensation factor ω_c in Eq. [10]. As long as $\omega > \omega_c$, compensation occurs, and $T_{act} = T_{pot}$. On the opposite, when $\omega_c > \omega$, compensation still occurs (if $\omega_c < 1$), but it does not fully compensate local reductions of water uptake. As a result, T_{act} is smaller than T_{pot}, which we call water stress. Jarvis (2011) showed that the α function can be derived from the soil matric flux potential function and the wilting point, while ω_c would be a plant attribute related to plant properties.

This formulation, however, has been subject to criticisms. Skaggs et al. (2006) exemplified that under uniform soil water content and root distributions, compensation would occur along the complete root profile, proportionally to potential water uptake, and would then result in keeping the default uptake distribution. This hypothetical example, in which compensation occurs without changing the uptake distribution, illustrates that definition of compensation by Jarvis (1989) may be in contradiction with the conceptual idea of compensation, which implies a spatial redistribution of water uptake. A second potential limitation of the above approach is the link between stress and compensation. Indeed, considering a case in which all soil water heads are in the flat part of the α function, no compensation can occur ($\omega = 1$), even when soil water head is heterogeneous, which should (according to the water flow equation applied to hydraulic networks) lead to decreased uptake in low water head zones compensated by increased uptake in high water head zones. Another problem arises from the fact that $\alpha(h)$ and ω_c are usually fitted from observed evolution in water content profiles while given water depletion profiles may correspond to several $\alpha(h)$ and ω_c leading to equifinality problems (Vandoorne et al., 2012).

Perspectives

Stress function and compensation processes are potentially coupled in two ways in macroscopic models. First, the Feddes stress function is sometimes used as a local soil water flow impedance function for defining compensation term (Eq. [9]). Second, compensation is conceptually linked to the stress onset. Stress and compensation should be decoupled as they originate from different processes; that is, stress is controlled by stomatal reduction of transpiration, while compensation is generated by the spatial distribution of soil and xylem water head. Equation [4] describes how local water uptake is controlled. It can be seen that the local uptake is made of two terms. The first term is the uncompensated part of the RWU, with a spatial distribution proportional to SUF and driven by T_{act}. The second term represents the compensation and depends on the difference between the local soil water head and the equivalent soil water head that is felt by the whole root system defined by Eq. [3]. The magnitude of compensation is controlled by the distribution of hydraulic properties within the root system through K_{comp} and SUF. As compared to Eq. [10], the main feature of Eq. [4] are that compensation and stress are fully decoupled and that compensation occurs as an additive term (independent of the T_{act}) rather than as a multiplicative term.

We used R-SWMS to model the root water uptake dynamics in a maize crop under drying conditions for 26 d and checked whether compensation, as defined in Eq. [10], could possibly fit simulations of a hydraulic architecture. To verify this, we calculated from Simulation 2a (Table 1) for each soil voxel and several times, the actual uptake, S_c, divided by the potential uptake, S_p, which according to Eq. [4] is defined as SUFT_{pot} so as to take into account the effect of root hydraulic properties on potential uptake. If Eq. [10] holds, this ratio should be equal to:

$$\frac{S_c(t)}{S_p(t)} = \frac{\alpha(h)}{\max[\omega(t), \omega_c]} \quad [11]$$

Equation [11] has the following properties: (i) its shape should be the same as that of α function (flat for high soil water heads and a linear increase with Hs for lower soil water heads), since $\max(\omega, \omega_c)$ is a constant at a given time, (ii) its intercept should stay the same at different times (generally, the wilting point is used as intercept), (iii) its slope for low soil water heads might depend on T_{pot}, (iv) its slope should depend on $\max(\omega, \omega_c)$, (v) its maximum should be higher [in case $\max(\omega, \omega_c) < 1$] or equal to 1 [in case $\max(\omega, \omega_c) = 1$], and (vi) its minimum should be 0.

Figure 6 presents the observed ratios between compensated and potential uptake vs. local soil water head at four selected times (corresponding to Subplots a, b, c and d), each point representing one root segment. Given our definition of the compensation, when the ratio is one, there is no compensation. Positive values higher or lower than 1 represent, respectively, higher or lower compensated uptake rates than under uniform water head. Negative values correspond to exudation, a particular state of compensation where water is released by roots. Subplots a (Day 2) and b (Day 20) were derived at noon, when T_{pot} is maximal, while Subplots c (Day 2) and d (Day 20) were derived in the evening, when T_{pot} is close to 0.

As predicted by Eq. [11], the ratio between compensated and potential uptake rates clearly show dependence to the local soil water head. However, this dependence does not have all of the expected properties. First, a linear decrease of the ratio is observed with decreasing soil water head, but no flat part can be seen for high water heads, even in wet conditions. This means that there should not be any plateau in the local soil impedance function in the macroscopic model. Second, the intercept is variable with time (e.g., −1000 cm on Fig. 6c, while −11000 cm in Fig. 6d). Third, the slope effectively shows a strong dependence to T_{pot}, with much higher values during low T_{pot} periods. Fourth, the product of the slope by T_{pot} is a constant during the whole scenario, which invalidates any dependence of the slope on $\max(\omega, \omega_c)$. Fifth, the maximum values of the ratio were always higher than 1, which means that $\max(\omega, \omega_c)$ would have always been lower than 1. Sixth, the minimum values of the ratio could be lower than 0, while S_c/S_p is positive by definition. This simulation clearly shows that Eq. [10] cannot reproduce the behavior of a hydraulic architecture model

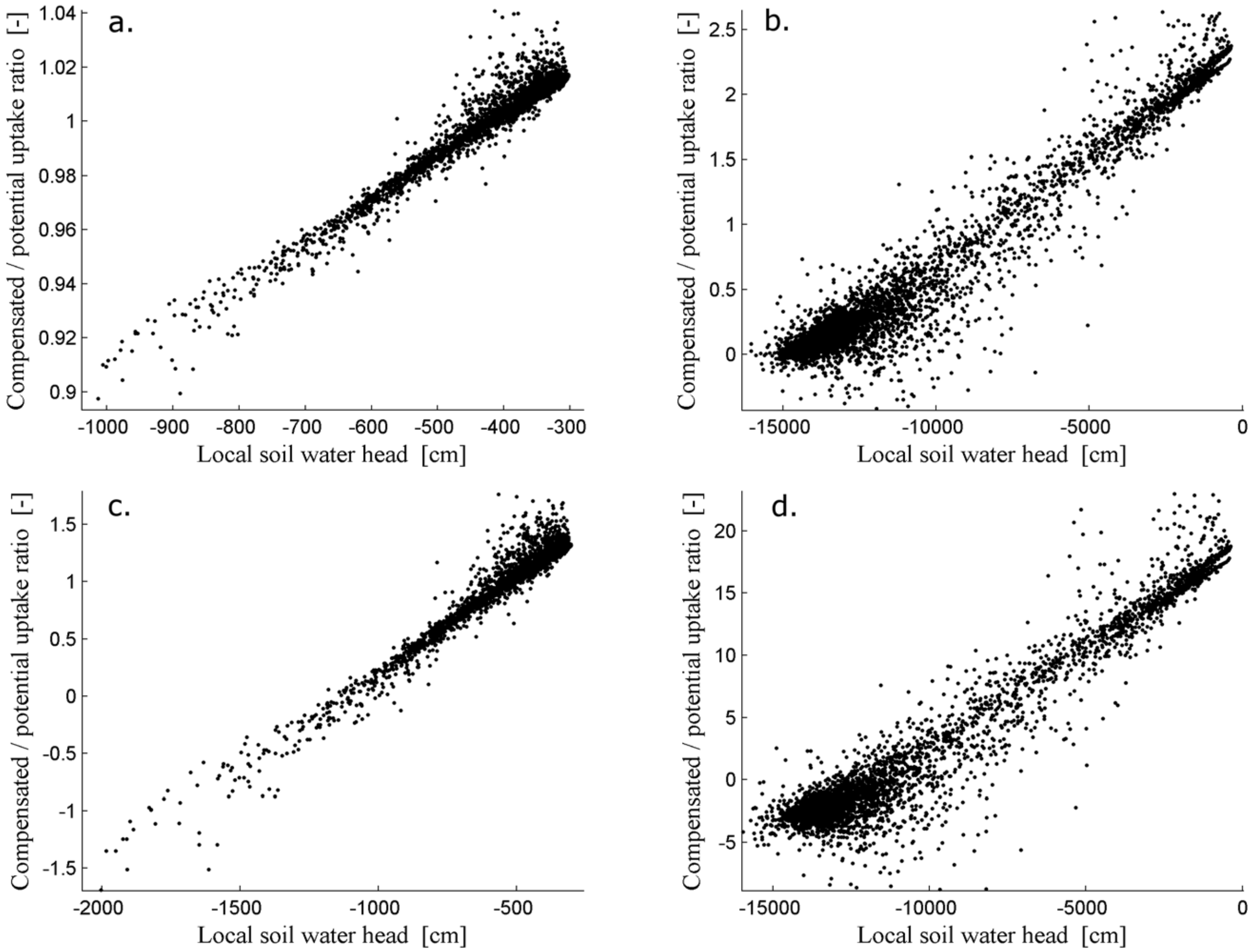


Fig. 6. Ratio between compensated and potential uptake rate for each root segment vs. local soil water head, at four specific times of scenario 2a. a. Day 2, noon (high T_{pot}); b. Day 20, noon (high T_{pot}); c. Day 2, evening (low T_{pot}); d. Day 20, evening (low T_{pot}).

under these conditions. It could however be checked whether a daily constant T_{pot} values and variable ω_c would improve the fit.

Conceptually, when root hydraulic resistance exists, it is intuitive that compensation (root hydraulic redistribution) should first depend on soil–root interface values (see Eq. [4]) and not bulk variables. Yet, as most macroscopic models do not explicitly simulate soil–root interface variables (and neglect root hydraulic resistance), their compensation function relies on soil hydraulic resistance depending on bulk soil properties (see Eq. [10]). This generates confusion in our understanding of the principles of the compensation process. We suggest that Eq. [4] is conceptually ideal for simulating root water uptake with compensation, but that additional physical soil hydraulic resistance functions should be used to link soil–root interface pressure head in Eq. [4] to bulk hydraulic variables, when needed.

Conclusions

We introduced a new perspective on modeling root water uptake based on a sink term approach by using a complex detailed numerical model and a simplified model. Root water uptake was reviewed by considering four main factors and the way these factors have been implemented in different macroscopic root water uptake models. It was shown that:

- Root resistance spatial distribution does not only depend on the root length distribution, but also on the hydraulic architecture. The root resistance may be well defined by the root length density but only for relatively wet soil and no limiting xylem conductance. For young root segments with low axial conductance or when embolism occurs, other approaches for describing root hydraulic architecture should be used, like the SUF proposed by Couvreur et al. (2012).
- Soil resistance needs to be accounted for in the sink term, in particular in the vicinity of the root surface (rhizosphere) where specific processes, such as root growth and root exudation, among others, may alter soil hydraulic properties as compared to the bulk soil and therefore affect root water uptake. Effective functions based on analytical solutions of the radial flow equation, such as Gardner (1960), exist to model water flow from bulk soil to root (e.g., de Jong van Lier et al., 2008). However, the impact of heterogeneous root, soil, and root pressure head distributions requires further consideration in such models.

- Compensation occurs as soon as there is a heterogeneous distribution of the soil water head around the root system. Locally, its magnitude is driven by the difference between the local water head at the soil–root interface and the root-averaged soil water head. Essentially, compensation is a function of the soil–root interface pressure head. Local soil impedance function should be used to link soil–root interface pressure head in Eq. [4] to bulk hydraulic variables.

- Stress and compensation are two different processes: stress is controlled by stomatal regulation responding to signals from roots in a drying soils, while compensation arises from non-uniform soil pressure head distribution in soil and roots. The modeling of these two processes should be definitely decoupled.

- Root activity for water uptake is not only controlled by root conductance properties. It is the result of the gradient of water head between soil root interface and xylem vessels, and the radial hydraulic properties of the root segment.

- The stress function should be defined as the function of an integrated or root–soil interface averaged pressure head rather than in terms of local bulk pressure heads. Since the pressure head losses are a function of the transpiration rate, it would also be better to define a maximal transpiration rate as a function of the soil–root or bulk soil pressure head rather than to define a ratio between actual and potential transpiration rates.

- In the simplest case, the hydraulic ability of the root system to sustain climatic demand is described by the root system equivalent conductance K_{rs} in Eq. [3]. Yet, nonlinearity in the stress function may arise from (i) nonisohydric stomatal regulation, (ii) the impact of nonlinear soil hydraulic conductivity function coupled with heterogeneous distribution of the uptake, and (iii) the temporal evolution of the root conductance (through air gap or physiological changes on K_r or K_x amongst others).

The simplified model proposed by Couvreur et al. (2012) (summarized in Eq. [2–4]) is an alternative approach to the traditional models for root water uptake when three-dimensional architecture has to be accounted for implicitly or explicitly. It has the advantage that it is based on physical concepts and it contains parameters which are potentially measurable like plant conductance and conductance distributions. Yet, the model does not explicitly consider the soil hydraulic resistance between bulk and soil–root interface. This can be accounted for by solving the Richards equation numerically or through analytical solutions. Note that similar mathematical expressions for root water uptake or compensation (corresponding with Eq. [4] but with different parameters) were found independently by other authors; see Eq. [9] in Volpe et al. (2013), Eq. [7] in Guswa (2012), and Eq. [17] in de Jong van Lier et al. (2013).

In this contribution, we presented simulation results of root water uptake using a detailed three-dimensional process model and we discussed how these results compare with concepts and functions for root water uptake that are used in macroscale models. The parameterization of the detailed process model may at a first seem a daunting task because of the numerous parameters that need to be defined. It should be noted, however, that some of these parameters such as root hydraulic properties and root architecture can be derived from direct measurements or observations. The detailed process model may be used to place the effect of processes and mechanisms that are studied in great detail in the plant science community, such as, plasticity of root system properties to environmental conditions (White et al., 2013) in context of a systemic soil–plant model (Draye et al., 2010). Other parameters such as the hydraulic properties of the rhizosphere and their difference to bulk soil properties are more difficult to derive directly. However, further development of noninvasive methods to characterize the structure of this interface provides promising prospects so as to observe processes and properties at the interface between soils and roots (Jury et al., 2011; Anderson and Hopmans, 2013). The combination of noninvasive methods that enable noninvasive monitoring of the root system, the moisture content distribution around roots and tracer transport towards roots with detailed process models may be used to validate and parameterize these types of model (Stingaciu et al., 2013). Validation of the process model does not necessarily require data with high spatial resolution. Monitoring of transpiration fluxes, soil moisture, water head, and root distributions (Garré et al., 2012) may as well provide information to constrain model parameters of the detailed model or of syntheses of the detailed model leading to simplified and physically based macroscopic models. Finally, the use of stable isotopes may be promising to derive directly root water uptake distributions, which are expected to constrain root water uptake models better than soil moisture and soil water head distributions (Durand et al., 2010).

Appendix

Definitions for resistance, resistivity, conductance, and conductivity, adapted from Hunt et al. (1991). Abbreviations: ψ, water pressure; l, length; H, water head.

	Flow (J) $[L^3T^{-1}]$		Flux density (q) $[LT^{-1}]$	
	Resistance	Conductance	Resistivity	Conductivity
Pressure gradient	$(d\psi/dl)/J$ $[T\,P\,L^{-4}]$	$J/(d\psi/dl)$ $[L^4\,T^{-1}\,P^{-1}]$	$(d\psi/dl)/q$ $[P\,T\,L^{-2}]$	$q/(d\psi/dl)$ $[L^2\,T^{-1}\,P^{-1}]$
Pressure difference	$\Delta\psi/J$ $[T\,P\,L^{-3}]$	$J/\Delta\psi$ $[L^3\,T^{-1}\,P^{-1}]$	$\Delta\psi/q$ $[P\,T\,L^{-1}]$	$q/\Delta\psi$ $[L\,T^{-1}\,P^{-1}]$
Head gradient	$(dH/dl)/J$ $[T\,L^{-3}]$	$J/(dH/dl)$ $[L^3\,T^{-1}]$	$(dH/dl)/q$ $[T\,L^{-1}]$	$q/(dH/dl)$ $[L\,T^{-1}]$
Head difference	$\Delta H/J$ $[T\,L^{-2}]$	$J/\Delta H$ $[L^2\,T^{-1}]$	$\Delta H/q$ $[T]$	$q/\Delta H$ $[T^{-1}]$

Acknowledgments

This work is a contribution of the Transregio Collaborative Research Center 32, Patterns in Soil–Vegetation–Atmosphere Systems: Monitoring, Modelling and Data Assimilation, which is funded by the German research association, DFG. V. Couvreur is supported by the "Fonds National de la Recherche Scientifique" (FNRS) of Belgium as a Research Fellow. The authors thank this funding agency for its financial support. The authors would also like to thank Nick Jarvis for his comments and suggestions.

References

Anderson, S.H., and J.W. Hopmans, editors. 2013. Soil–water–root processes: Advances in tomography and imaging. SSSA Spec. Publ. 61. SSSA, Madison, WI.

Alm, D.M., J. Cavelier, and P.S. Nobel. 1992. A finite-element model of radial and axial conductivities for individual roots—Development and validation for 2 desert succulents. Ann. Bot. (Lond.) 69:87–92.

Aravena, J.E., M. Berli, T.A. Ghezzehei, and S.W. Tyler. 2011. Effects of root-induced compaction on rhizosphere hydraulic properties—X-ray microtomography imaging and numerical simulations. Environ. Sci. Technol. 45(2):425–431. doi:10.1021/es102566j

Beff, L., T. Günther, B. Vandoorne, V. Couvreur, and M. Javaux. 2013. Three-dimensional monitoring of soil water content in a maize field using electrical resistivity tomography. Hydrol. Earth Syst. Sci. 17:595–609. doi:10.5194/hess-17-595-2013

Betts, R.A. 2005. Integrated approaches to climate-crop modelling: Needs and challenges. Philos. Trans. R. Soc. B. Biol. Sci. 360(1463):2049–2065. doi:10.1098/rstb.2005.1739

Carminati, A., and D. Vetterlein. 2012. Plasticity of rhizosphere hydraulic properties as a key for efficient utilization of scarce resources. Ann. Bot. (Lond.) 112(2):277–290. doi:10.1093/aob/mcs262

Carminati, A., D. Vetterlein, U. Weller, H.J. Vogel, and S.E. Oswald. 2009. When roots lose contact. Vadose Zone J. 8:805–809. doi:10.2136/vzj2008.0147

Carminati, A. 2013. Rhizosphere wettability decreases with root age: A problem or a strategy to increase water uptake of young roots? Front. Plant Sci. 4. doi:10.3389/fpls.2013.00298

Carsel, R.F., and R.S. Parrish. 1988. Developing joint probability distributions of soil water retention characteristics. Water Resour. Res. 24:755–769. doi:10.1029/WR024i005p00755

Couvreur, V., J. Vanderborght, and M. Javaux. 2012. A simple three-dimensional macroscopic root water uptake model based on the hydraulic architecture approach. Hydrol. Earth Syst. Sci. 16:2957–2971 doi:10.5194/hess-16-2957-2012

Comstock, J., and M. Mencuccini. 1998. Control of stomatal conductance by leaf water potential in *Hymenoclea salsola* (T. & G.), a desert subshrub. Plant Cell Environ. 21(10):1029–1038. doi:10.1046/j.1365-3040.1998.00353.x

Cowan, I.R. 1965. Transport of water in the soil-plant-atmosphere system. J. Appl. Ecol. 2:221–239. doi:10.2307/2401706

Damour, G., T. Simonneau, H. Cochard, and L. Urban. 2010. An overview of models of stomatal conductance at the leaf level. Plant Cell Environ. 33(9):1419–1438. doi:10.1111/j.1365-3040.2010.02181.x.

de Jong van Lier, Q., J.C. van Dam, K. Metselaar, R. de Jong, and W.H.M. Duijnisveld. 2008. Macroscopic root water uptake distribution using a matric flux potential approach. Vadose Zone J. 7:1065–1078. doi:10.2136/vzj2007.0083

de Jong van Lier, Q., J.C. van Dam, A. Durigon, M.A. dos Santos, and K. Metselaar. 2013. Modeling water potentials and flows in the soil–plant system comparing hydraulic resistances and transpiration reduction functions. Vadose Zone J. 12. doi:10.2136/vzj2013.02.0039.

de Willigen, P., J.C. van Dam, M. Javaux, and M. Heinen. 2012. Root water uptake as simulated by three soil water flow models. Vadose Zone J. 11(3). doi:10.2136/vzj2012.0018

Dodd, I.C., G. Egea, C.W. Watts, and W.R. Whalley. 2010. Root water potential integrates discrete soil physical properties to influence ABA signalling during partial rootzone drying. J. Exp. Bot. 61(13):3543–3551. doi:10.1093/jxb/erq195

Doussan, C., L. Pagès, and G. Vercambre. 1998a. Modelling of the hydraulic architecture of root systems: An integrated approach to water absorption—Model description. Ann. Bot. (Lond.) 81(2):213–223. doi:10.1006/anbo.1997.0540

Doussan, C., G. Vercambre, and L. Pagès. 1998b. Modelling of the hydraulic architecture of root systems: An integrated approach to water absorption—Distribution of axial and radial conductances in maize. Ann. Bot. (Lond.) 81(2):225–232. doi:10.1006/anbo.1997.0541

Doussan, C., A. Pierret, E. Garrigues, and L. Pages. 2006. Water uptake by plant roots: II. Modelling of water transfer in the soil root-system with explicit account of flow within the root system—Comparison with experiments. Plant Soil 283(1–2):99–117. doi:10.1007/s11104-004-7904-z

Draye, X., Y. Kim, G. Lobet, and M. Javaux. 2010. Model-assisted integration of physiological and environmental constraints affecting the dynamic and spatial patterns of root water uptake from soils. J. Exp. Bot. 61(8):2145–2155. doi:10.1093/jxb/erq077

Durand, J.L., T. Bariac, Y. Rothfuss, P. Richard, P. Biron, and F. Gastal. 2010. Investigating the competition for water and the depth of water extraction in multispecies grasslands using ^{18}O natural abundance. In: C. Huyghe, editor, Sustainable use of genetic diversity in forage and turf breeding, Part 3. Springer, New York, p. 205–209.

Faria, L.N., M.G. Rocha, Q. Jong Van Lier, and D. Casaroli. 2009. A split-pot experiment with sorghum to test a root water uptake partitioning model. Plant Soil 331(1–2):299–311. doi:10.1007/s11104-009-0254-0

Feddes, R.A., P. Kowalik, K. Kolinska-Malinka, and H. Zaradny. 1976. Simulation of field water uptake by plants using a soil water dependent root extraction function. J. Hydrol. 31(1–2):13–26. doi:10.1016/0022-1694(76)90017-2

Feddes, R.A., P.J. Kowalik, and H. Zaradny. 1978. Simulation of field water use and crop yield. John Wiley & Sons, New York.

Fiscus, E.L. 1977. Effects of coupled solute and water-flow in plant roots with special reference to Brouwers experiment. J. Exp. Bot. 28(1):71–77 10.1093/jxb/28.1.71. doi:10.1093/jxb/28.1.71

Gardner, W.R. 1960. Dynamic aspects of water availability to plants. Soil Sci. 89:63–73. doi:10.1097/00010694-196002000-00001

Garré, S., E. Laloy, L. Pagès, M. Javaux, J. Vanderborght, and H. Vereecken. 2012. Parameterizing the root system development of spring barley using minirhizotron data. Vadose Zone J. 11(4). doi:10.2136/vzj2011.0179.

Gradmann, H. 1928. Untersuchungen Über Die Wasserverhältnisse Des Bodens Als Grundlage Des Pflanzenwachstums. I. Jahrb. wiss. Bot. 69:1–100.

Guswa, A.J. 2012. Canopy versus roots: Production and destruction of variability in soil moisture and hydrologic fluxes. Vadose Zone J. 11(3). doi:10.2136/vzj2011.0159

Hachez, Ch., D. Veselov, Q. Ye, H. Reinhardt, T. Knipfer, W. Fricke, and F. Chaumont. 2012. Short-term control of maize cell and root water permeability through plasma membrane aquaporin isoforms. Plant Cell Environ. 35(1):185–198. doi:10.1111/j.1365-3040.2011.02429.x

Herkelrath, W.N., E.E. Miller, and W.R. Gardner. 1977. Water uptake by plants: II. The root contact model. Soil Sci. Soc. Am. J. 41(6):1039–1043. doi:10.2136/sssaj1977.03615995004100060004x

Hinsinger, P., G.R. Gobran, P.J. Gregory, and W.W. Wenzel. 2005. Rhizosphere geometry and heterogeneity arising from root-mediated physical and chemical processes. New Phytol. 168(2):293–303. doi:10.1111/j.1469-8137.2005.01512.x

Hodge, A., G. Berta, C. Doussan, F. Merchan, and M. Crespi. 2009. Plant root growth, architecture and function. Plant Soil 321(1–2):153–187. doi:10.1007/s11104-009-9929-9

Hopmans, J.W., and K.L. Bristow. 2002. Current capabilities and future needs of root water and nutrient uptake modeling. Adv. Agron. 77:103–183. doi:10.1016/S0065-2113(02)77014-4

Hose, E., E. Steudle, and W. Hartung. 2000. Abscisic acid and hydraulic conductivity of maize roots: A study using cell- and root-pressure probes. Planta 211(6):874–882. doi:10.1007/s004250000412

Hunt, E., S. Running, and C. Federer. 1991. Extrapolating plant water-flow resistances and capacitances to regional scales. Agric. For. Meteorol. 54(2–4):169–195. doi:10.1016/0168-1923(91)90005-B

Jarvis, N.J. 1989. A simple empirical-model of root water-uptake. J. Hydrol. 107(1–4):57–72. doi:10.1016/0022-1694(89)90050-4

Jarvis, N.J. 2011. Simple physics-based models of compensatory plant water uptake: Concepts and eco-hydrological consequences. Hydrol. Earth Syst. Sci. 15(11):3431–3446. doi:10.5194/hess-15-3431-2011

Javaux, M., T. Schroeder, J. Vanderborght, and H. Vereecken. 2008. Use of a three-dimensional detailed modeling approach for predicting root water uptake. Vadose Zone J. 7(3):1079–1088. doi:10.2136/vzj2007.0115

Janott, M., S. Gayler, A. Gessler, M. Javaux, C. Klier, and E. Priesack. 2011. A one-dimensional model of water flow in soil-plant systems based on plant architecture. Plant Soil 341:233–256. doi:10.1007/s11104-010-0639-0

Javot, H., and C. Maurel. 2002. The role of aquaporins in root water uptake. Ann. Bot. (Lond.) 90(3):301–313. doi:10.1093/aob/mcf199

Jury, W.A., D. Or, Y. Pachepsky, H. Vereecken, J.W. Hopmans, L.R. Ahuja, B.E. Clothier, K.L. Bristow, G.J. Kluitenberg, P. Moldrup, J. Šimůnek, M.Th. van Genuchten, and R. Horton. 2011. Kirkham's legacy and contemporary challenges in soil physics research. Soil Sci. Soc. Am. J. 75(5):1589. doi:10.2136/sssaj2011.0115

Katul, G., P. Todd, D. Pataki, Z.J. Kabala, and R. Oren. 1997. Soil water depletion by oak trees and the influence of root water uptake on the moisture content spatial statistics. Water Resour. Res. 33(4):611–623. doi:10.1029/96WR03978

Knipfer, T., and W. Fricke. 2010. Root pressure and a solute reflection coefficient close to unity exclude a purely apoplastic pathway of radial water transport in barley (*Hordeum vulgare*). New Phytol. 187(1):159–170. doi:10.1111/j.1469-8137.2010.03240.x

Landsberg, J.J., and N.D. Fowkes. 1978. Water-movement through plant roots. Ann. Bot. (Lond.) 42:493–508. doi:10.1093/oxfordjournals.aob.a085488

Lehmann, J. 2003. Subsoil root activity in tree-based cropping systems. Plant Soil 255:319–331. doi:10.1023/A:1026195527076

Lhomme, J.p. 1998. Formulation of root water uptake in a multi-layer soil-plant model: Does van den Honert's equation hold? Hydrol. Earth Syst. Sci. 2:31–40. doi:10.5194/hess-2-31-1998

Li, K.Y., R. de Jong, and J.B. Boisvert. 2001. An exponential root-water-uptake model with water stress compensation. J. Hydrol. 252(1–4):189–204. doi:10.1016/S0022-1694(01)00456-5

Lobet, G., Ch. Hachez, M. Javaux, and X. Draye. 2013. Root water uptake and water flow in the soil-root domain. In: A. Eshel and T. Beeckman, editors, Plant roots: The hidden half. 4th ed. CRC Press, Boca Raton, FL.

Maurel, C., and M.J. Chrispeels. 2001. Aquaporins. A molecular entry into plant water relations. Plant Physiol. 125(1):135–138. doi:10.1104/pp.125.1.135

Maurel, C., T. Simonneau, and M. Sutka. 2010. The significance of roots as hydraulic rheostats. J. Exp. Bot. 61(12):3191–3198. doi:10.1093/jxb/erq150

Molz, F.J. 1976. Water transport in the soil-root system: Transient analysis. Water Resour. Res. 12(4):805–808. doi:10.1029/WR012i004p00805

Molz, F.J. 1981. Models of water transport in the soil-plant system: A review. Water Resour. Res. 17(5):1245–1260. doi:10.1029/WR017i005p01245

Moradi, A.B., A. Carminati, A. Lamparter, S.K. Woche, J. Bachmann, D. Vetterlein, H.-J. Vogel, and S.E. Oswald. 2012. Is the rhizosphere temporarily water repellent? Vadose Zone J. doi:10.2136/vzj2011.0120.

Nimah, M.N., and R.J. Hanks. 1973. Model for estimating soil water, plant, and atmospheric interrelations: I. Description and sensitivity. Soil Sci. Soc. Am. J. 37:522. doi:10.2136/sssaj1973.03615995003700040018x

Oki, T., and S. Kanae. 2006. Global hydrological cycles and world water resources. Science 313(5790):1068–1072. doi:10.1126/science.1128845

Pages, L., G. Vercambre, J.L. Drouet, F. Lecompte, C. Collet, and J. Le Bot. 2004. Root Typ: A generic model to depict and analyse the root system architecture. Plant Soil 258(1):103–119. doi:10.1023/B:PLSO.0000016540.47134.03

Pierret, A., C. Doussan, Y. Capowiez, F. Bastardie, and L. Pagès. 2007. Root functional architecture: A framework for modeling the interplay between roots and soil. Vadose Zone J. 6(2):269. doi:10.2136/vzj2006.0067

Philip, J.R. 1966. Plant water relations—Some physical aspects. Annu. Rev. Plant Physiol. 17:245–268. doi:10.1146/annurev.pp.17.060166.001333

Porporato, A., P. D'odorico, F. Laio, L. Ridolfi, and I. Rodriguez-Iturbe. 2002. Ecohydrology of water-controlled ecosystems. Adv. Water Resour. 25(8):1335–1348. doi:10.1016/S0309-1708(02)00058-1

Prieto, I., C. Armas, and F.I. Pugnaire. 2012. Water release through plant roots: New insights into its consequences at the plant and ecosystem level. New Phytol. 193(4):830–841. doi:10.1111/j.1469-8137.2011.04039.x

Schneider, C.L., S. Attinger, J.O. Delfs, and A. Hildebrandt. 2010. Implementing small scale processes at the soil-plant interface-the role of root architectures for calculating root water uptake profiles. Hydrol. Earth. Syst. Sci. 14(2):279–289. doi:10.5194/hess-14-279-2010

Schröder, T., M. Javaux, J. Vanderborght, B. Koerfgen, and H. Vereecken. 2009. Implementation of a microscopic soil-root hydraulic conductivity drop function in a three-dimensional soil-root architecture water transfer model. Vadose Zone J. 8(3):783–792. doi:10.2136/vzj2008.0116

Schröder, N., M. Javaux, J. Vanderborght, B. Steffen, and H. Vereecken. 2012. Effect of root water and solute uptake on apparent soil dispersivity: A simulation study. Vadose Zone J. 11(3). doi:10.2136/vzj2012.0009.

Seneviratne, S.I., D. Luethi, M. Litschi, and Ch. Schaer. 2006. Land-atmosphere coupling and climate change in Europe. Nature 443(7108):205–209. doi:10.1038/nature05095

Šimůnek, J., and J.W. Hopmans. 2009. Modeling compensated root water and nutrient uptake. Ecol. Modell. 220(4):505–521. doi:10.1016/j.ecolmodel.2008.11.004

Simunek, J., K. Huang, and M.T. van Genuchten. 1995. The SWMS-3D Code for simulating water flow and solute transport in three-dimensional variably-saturated media. Res. Rep.139. U.S. Salinity Laboratory, Riverside, CA.

Skaggs, T.H., M.Th. van Genuchten, P.J. Shouse, and J.A. Poss. 2006. Macroscopic approaches to root water uptake as a function of water and salinity stress. Agric. Water Manage. 86:140–149. doi:10.1016/j.agwat.2006.06.005

So, H.B., L.A.G. Aylmore, and J.P. Quirk. 1976. The resistance of intact maize roots to water flow. Soil Sci. Soc. Am. J. 40:222–225. doi:10.2136/sssaj1976.03615995004000020012x

Sperry, J.S., U.G. Hacke, R. Oren, and J.P. Comstock. 2002. Water deficits and hydraulic limits to leaf water supply. Plant Cell Environ. 25(2):251–263. doi:10.1046/j.0016-8025.2001.00799.x

Sperry, J.S., U.G. Hacke, and J.K. Wheeler. 2005. Comparative analysis of end wall resistivity in xylem conduits. Plant Cell Environ. 28(4):456–465. doi:10.1111/j.1365-3040.2005.01287.x

Steudle, E., and C.A. Peterson. 1998. How does water get through roots? J. Exp. Bot. 49(322):775–788. doi:10.1093/jexbot/49.322.775.

Stingaciu, L., H. Schulz, A. Pohlmeier, S. Behnke, H. Zilken, M. Javaux, and H. Vereecken. 2013. In situ root system architecture extraction from magnetic resonance imaging for water uptake modeling. Vadose Zone J. 12(1): doi:10.2136/vzj2012.0019.

Tardieu, F. 1996. Drought perception by plants do cells of droughted plants experience water stress? Plant Growth Regul. 20(2):93–104. doi:10.1007/BF00024005

Tardieu, F., and W.J. Davies. 1993. Integration of hydraulic and chemical signalling in the control of stomatal conductance and water status of droughted plants. Plant Cell Environ. 16(4):341–349. doi:10.1111/j.1365-3040.1993.tb00880.x

Taylor, S.A., and M.G. Ashcroft. 1972. Physical edaphology. Freeman and Co., San Francisco, CA.

Teuling, A.J., R. Uijlenhoet, F. Hupet, and P.A. Troch. 2006. Impact of plant water uptake strategy on soil moisture and evapotranspiration dynamics during drydown. Geophys. Res. Lett. 33(3). doi:10.1029/2005GL025019

Thévenin, L. 1883. Extension of Ohm's law to complex electromotive circuits. Ann. Télégraph. 10:222–224.

Tuzet, A., A. Perrier, and R. Leuning. 2003. A coupled model of stomatal conductance, photosynthesis and transpiration. Plant Cell Environ. 26(7):1097–1116. doi:10.1046/j.1365-3040.2003.01035.x

Van den Honert, T.H. 1948. Water transport in plants as a catenary process. Discuss. Faraday Soc. 3:146–153. doi:10.1039/df9480300146

Vandoorne, B., L. Beff, S. Lutts, and M. Javaux. 2012. Root water uptake dynamics of Cichorium intybus var. sativum under water-limited conditions. Vadose Zone J. 11(3). doi:10.2136/vzj2012.0005

Vereecken, H., P. Burauel, J. Groeneweg, E. Klumpp, W. Mittelstaedt, H.-D. Narres, T. Pütz, J. van der Kruk, J. Vanderborght, and F. Wendland. 2009. Research at the agrosphere institute: From the process scale to the catchment scale. Vadose Zone J. 8(3):664. doi:10.2136/vzj2008.0143

Volpe, V., M. Marani, J.D. Albertson, and G. Katul. 2013. Root controls on water redistribution and carbon uptake in the soil-plant system under current and future climate. Adv. Water Resour. 60:110–120. doi:10.1016/j.advwatres.2013.07.008

Vrugt, J.A., M.T. Van Wijk, J.W. Hopmans, and J. Simunek. 2001. One-, two-, and three-dimensional root water uptake functions for transient modeling. Water Resour. Res. 37(10):2457–2470. doi:10.1029/2000WR000027

Wesseling, J.G., J.A. Elbers, P. Kabat, and B.J. van den Broek. 1991. SWATRE: Instructions for input, internal note. Winand Staring Centre, Wageningen, the Netherlands.

White, Ph.J., T.S. George, P.J. Gregory, A.G. Bengough, P.D. Hallett, and B.M. McKenzie. 2013. Matching roots to their environment. Ann. Bot. (Lond.) 112:207–222. doi:10.1093/aob/mct123